Warehouse Inventory System

Project Report submitted for the partial fulfillment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
V Sameer Reddy (2520030276)
Vennela Edala (2520030288)
Ch Sahasra Gayathri (2520030306)

Team Number: 17
Section: 9

Under the guidance of

Dr. S. Vinay Kumar, Associate Professor,
Department of Computer Science and Engineering

(DEEMED TO BE UNIVERSITY)
BACHUPALLY
Hyderabad-500090, Telangana, India.

ABSTRACT

The **Warehouse Inventory System** is designed to improve the efficiency and accuracy of inventory management in warehouses handling large volumes of products. Manual inventory processes can make it difficult to locate products, monitor stock levels, maintain supplier information, and update records efficiently. This project provides a structured solution by applying fundamental **Data Structures and Algorithms (DSA)** concepts to warehouse operations.

The system enables users to add, update, delete, search, and view products, identify low-stock items, manage supplier details, and generate inventory reports. Data structures such as **ArrayList** and **HashMap**, along with algorithms including **Binary Search, Merge Sort, and linear traversal**, are used for efficient data storage, searching, sorting, and retrieval. The system is developed using **Java**, **MySQL**, and **JDBC** for database connectivity. Overall, the project aims to reduce manual effort, minimize errors, and provide a faster and more organized approach to warehouse inventory management.

Signature of the Guide